

Below are **100+ Java Lab Tasks/Exercises** based on the **24ITT32 – Programming in Java** syllabus, arranged from basic to advanced.

UNIT I – Classes and Objects (25 Tasks)

1. Print "Hello World" using Java.
2. Perform arithmetic operations.
3. Find largest of three numbers.
4. Check whether a number is even or odd.
5. Find factorial of a number.
6. Generate Fibonacci series.
7. Check palindrome number.
8. Reverse a number.
9. Sum of digits of a number.
10. Check Armstrong number.
11. Find prime number.
12. Generate prime numbers between ranges.
13. Matrix addition.
14. Matrix multiplication.
15. Transpose of matrix.
16. Sorting array in ascending order.
17. Search an element in array.
18. Create Student class.
19. Create Employee class.
20. Calculate area using class methods.
21. Demonstrate object creation.
22. Demonstrate method calling.
23. Create Book class.
24. Create Bank Account class.
25. Student Result Processing System.

UNIT II – Constructors and Nested Classes (20 Tasks)

26. Default constructor example.
27. Parameterized constructor example.
28. Constructor overloading.
29. Copy constructor simulation.
30. Demonstrate `this` keyword.

31. Demonstrate garbage collection.
32. Method overloading.
33. Recursive factorial.
34. Recursive Fibonacci.
35. Pass object as argument.
36. Return object from method.
37. Static variable example.
38. Static method example.
39. Command line argument addition.
40. Command line calculator.
41. Variable-length argument example.
42. Nested class demonstration.
43. Inner class example.
44. Student Management using constructors.
45. Product Inventory System.

UNIT III – Inheritance, Packages and Interfaces (20 Tasks)

46. Single inheritance.
47. Multilevel inheritance.
48. Hierarchical inheritance.
49. Method overriding.
50. Dynamic method dispatch.
51. Abstract class implementation.
52. Abstract method example.
53. Final variable demonstration.
54. Final method demonstration.
55. Final class demonstration.
56. Package creation.
57. User-defined package.
58. Import package example.
59. Interface implementation.
60. Multiple interfaces.
61. Interface inheritance.
62. Shape calculation using interfaces.
63. Employee Payroll using inheritance.
64. Vehicle Management System.
65. Hospital Management using inheritance.

UNIT IV – Exception Handling, Multithreading and I/O (20 Tasks)

- 66. Try-Catch example.
- 67. Multiple Catch blocks.
- 68. Nested Try blocks.
- 69. Finally block example.
- 70. Throw keyword example.
- 71. Throws keyword example.
- 72. User-defined exception.
- 73. Bank insufficient balance exception.
- 74. Divide by zero exception handling.
- 75. File not found exception.
- 76. Create thread using Thread class.
- 77. Create thread using Runnable interface.
- 78. Multiple thread execution.
- 79. Thread priority demonstration.
- 80. Thread synchronization.
- 81. Producer Consumer simulation.
- 82. Read data from keyboard.
- 83. Write data to file.
- 84. Read data from file.
- 85. Student Record File Management System.

UNIT V – String Handling, Generics and Collections (25 Tasks)

- 86. String comparison.
- 87. String concatenation.
- 88. String reversal.
- 89. Count vowels in string.
- 90. Check palindrome string.
- 91. StringBuffer example.
- 92. StringBuilder example.
- 93. Find duplicate characters.
- 94. Word count program.
- 95. Generic class example.
- 96. Generic method example.
- 97. Generic interface example.
- 98. ArrayList operations.

- 99. LinkedList operations.
- 100. HashSet operations.
- 101. TreeSet operations.
- 102. HashMap operations.
- 103. TreeMap operations.
- 104. Vector example.
- 105. Stack implementation.
- 106. Queue implementation.
- 107. Collection sorting.
- 108. Student Record using ArrayList.
- 109. Employee Management using HashMap.
- 110. Library Management System using Collections.

Mini Projects (10)

- 111. Student Record Management System
- 112. Library Management System
- 113. Banking Management System
- 114. Hospital Management System
- 115. Online Quiz Application
- 116. Inventory Management System
- 117. Employee Payroll System
- 118. Hotel Reservation System
- 119. Contact Management System
- 120. Course Registration System

These **120 Java tasks** cover the complete syllabus and are suitable for **semester lab, assignments, record work, internal assessment, and mini projects.**